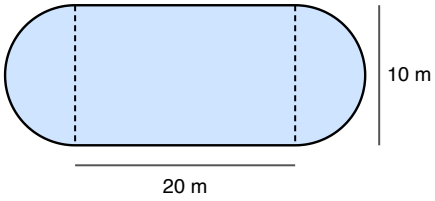
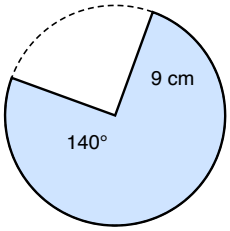
Shape and Space 7 — Circles and sectors · Solids · Similar shapes

Calculator allowed. Marks in brackets. ★ = harder question. Give answers to 3 s.f. unless exact. Area of a circle πr^2, circumference $2\pi r$.

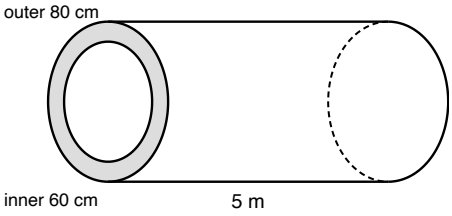
1 A running track is a rectangle 20 m by 10 m with a semicircle on each 10 m end. Find its perimeter and its area. [4]



2 A sector of a circle of radius 9 cm has angle 140°. Find its area and its perimeter. [4]



3 A hollow concrete pipe is 5 m long. Its outer diameter is 80 cm and its inner diameter is 60 cm. Find the volume of concrete in m³. [5]



4 Two bottles are similar. The smaller is 12 cm tall and the larger 18 cm tall. [6]

a The surface area of the smaller bottle is 150 cm². Find the surface area of the larger.

b The volume of the larger bottle is 810 cm³. Find the volume of the smaller.

5 Cone A is 20 cm tall with volume 600 cm³ and curved surface area 320 cm². Cone B is similar to A and has volume 1500 cm³. [6]

a Find the height of cone B. **b** Find the curved surface area of cone B.

Total: 25 marks

Check: sector: $\frac{\theta}{360}$ of the area or of the circumference; perimeter adds two radii · prism/cylinder: cross-section × length · hollow shapes: subtract areas, then multiply by length · similar shapes: lengths ×k, areas ×k^2, volumes × k^3; from a volume ratio take the cube root to get k.